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Agrément Certificate
24/7275
Product Sheet 1 Issue 1

ALUMASC EUROROOF BITUMINOUS ROOFING MEMBRANES

EUROROOF MONO SYSTEM

This Agrément Certificate Product Sheet(1) relates to the Euroroof Mono System, a polyester reinforced, SBS modified bituminous membrane and air and vapour control layers for use as a fully or partially adhered, single layer waterproofing system on new and existing buildings. The system is for use on pitched, flat and protected roofs, and podia, with limited access or pedestrian access with suitable protection.

(1) Hereinafter referred to as 'Certificate'.

The assessment includes

Product factors:
- compliance with Building Regulations
- compliance with additional regulatory or non-regulatory information where applicable
- evaluation against technical specifications
- assessment criteria and technical investigations
- uses and design considerations

Process factors:
- compliance with Scheme requirements
- installation, delivery, handling and storage
- production and quality controls
- maintenance and repair

Ongoing contractual Scheme elements†:
- regular assessment of production
- formal 3-yearly review

KEY FACTORS ASSESSED
Section 1. Mechanical resistance and stability
Section 2. Safety in case of fire
Section 3. Hygiene, health and the environment
Section 4. Safety and accessibility in use
Section 5. Protection against noise
Section 6. Energy economy and heat retention
Section 7. Sustainable use of natural resources
Section 8. Durability

The BBA has awarded this Certificate to the company named above for the system described herein. This system has been assessed by the BBA as being fit for its intended use provided it is installed, used and maintained as set out in this Certificate.

On behalf of the British Board of Agrément

Date of issue: 19 November 2024

Hardy Giesler
Chief Executive Officer

This BBA Agrément Certificate is issued under the BBA's Inspection Body accreditation to ISO/IEC 17020. Sections marked with † are not issued under accreditation.

The BBA is a UKAS accredited Inspection Body (No. 4345), Certification Body (No. 0113) and Testing Laboratory (No. 0357).

Readers MUST check that this is the latest issue of this Agrément Certificate by either referring to the BBA website or contacting the BBA directly.

The Certificate should be read in full as it may be misleading to read clauses in isolation.

Any photographs are for illustrative purposes only, do not constitute advice and should not be relied upon.

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SUMMARY OF ASSESSMENT AND COMPLIANCE

This section provides a summary of the assessment conclusions; readers should refer to the later sections of this Certificate for information about the assessments carried out.

Compliance with Regulations

Having assessed the key factors, the opinion of the BBA is that the Euroroof Mono System, if installed, used and maintained in accordance with this Certificate, can satisfy or contribute to satisfying the relevant requirements of the following Building Regulations:

The Building Regulations 2010 (England and Wales) (as amended)

Requirement: B4(1) External fire spread
Comment: The system is restricted by this Requirement in some circumstances. See section 2 of this Certificate.

Requirement: B4(2) External fire spread
Comment: On a suitable substructure, the system may enable a roof to be unrestricted under this Requirement. See section 2 of this Certificate.

Requirement: C2(b) Resistance to moisture
Comment: The system, including joints, will enable a roof to satisfy this Requirement. See section 3 of this Certificate.

Requirement: C2(c) Resistance to moisture
Comment: The system, including joints, can contribute to a roof to satisfying this Requirement. See section 3 of this Certificate.

Regulation: 7(1) Materials and workmanship
Comment: The system is acceptable. See sections 8 and 9 of this Certificate.

The Building (Scotland) Regulations 2004 (as amended)

Regulation: 8(1)(2) Fitness and durability of materials and workmanship
Comment: The use of the system satisfies this Regulation. See sections 8 and 9 of this Certificate.

Regulation: 9 Building standards - construction
Standard: 2.8 Spread from neighbouring buildings
Comment: The system, when applied to a suitable substructure, may enable a roof to be unrestricted by this Standard with reference to clause 2.8.1[1][2]. See section 2 of this Certificate.

Standard: 3.10 Precipitation
Comment: The system, including joints, will enable a roof to satisfy this Standard, with reference to clauses 3.10.1[1][2] and 3.10.7[1][2]. See section 3 of this Certificate.

Standard: 7.1(a) Statement of sustainability
Comment: The system can contribute to satisfying the relevant requirements of Regulation 9, Standards 1 to 6, and therefore will contribute to a construction meeting a bronze level of sustainability as defined in this Standard.

Standard: 3.15 Condensation
Comment: The system will enable a roof to satisfy this Standard, with reference to clauses 3.15.1[1], 3.15.3[1], 3.15.5[1] and 3.15.6[1]. See section 3 of this Certificate.

Regulation:	**12**	**Building standards - conversion**
Comment:		Comments in relation to the system under Regulation 9, Standards 1 to 6, also apply to this Regulation, with reference to clause 0.12.1[(1)(2)] and Schedule 6[(1)(2)].
		(1) Technical Handbook (Domestic). (2) Technical Handbook (Non-Domestic).

The Building Regulations (Northern Ireland) 2012 (as amended)

Regulation:	**23(1)(a)(i)**	**Fitness of materials and workmanship**
Comment:	**(iii)(b)(i)**	The system can satisfy this Regulation. See sections 8 and 9 of this Certificate.
Regulation:	**28(b)**	**Resistance to moisture and weather**
Comment:		The system can satisfy this Regulation. See section 3 of this Certificate.
Regulation:	**29**	**Condensation**
Comment:		The system can contribute to a roof satisfying this Regulation. See section 3 of this Certificate.
Regulation:	**36(a)**	**External fire spread**
Comment:		The system is restricted by this Regulation in some circumstances. See section 2 of this Certificate.
Regulation:	**36(b)**	**External fire spread**
Comment:		On a suitable substructure, the system may enable a roof to be unrestricted under this Regulation. See section 2 of this Certificate.

Additional Information

NHBC Standards 2024

In the opinion of the BBA, the Euroroof Mono System, if installed, used and maintained in accordance with this Certificate, can satisfy or contribute to satisfying the relevant requirements in relation to *NHBC Standards*, Chapter 7.1 *Flat roofs, terraces and balconies*.

In addition, in the opinion of the BBA, the system, when installed and used in accordance with this Certificate, can satisfy or contribute to satisfying the relevant requirements in relation to *NHBC Standards for Conversions and Renovations*, taking account of other relevant guidance within the chapter and the suitability of the substrate to receive the system.

The NHBC Standards do not cover the refurbishment of existing roofs.

Fulfilment of Requirements

The BBA has judged the Euroroof Mono System to be satisfactory for use as described in this Certificate. The system has been assessed as a fully or partially adhered, single layer waterproofing system on new and existing buildings. The system is for use on pitched, flat and protected roofs, and podia, with limited access or pedestrian access with suitable protection.

ASSESSMENT

Product description and intended use

The Certificate holder provided the following description for the system under assessment. The Euroroof Mono System consists of:

- Euroroof Mono Cap Sheet— a styrene-butadiene-styrene (SBS) modified bitumen membrane, with a stabilised polyester reinforcement and mineral finish in various colours
- Euroroof PU Membrane Adhesive — a single component, low foaming polyurethane adhesive
- Euroroof Torch-On AVCL— a polyester reinforced, torch-on, SBS modified bitumen vapour barrier with an aluminium core, with a polyethylene film on the underside and a sanded upper surface
- Euroroof Self-Adhesive AVCL — a polyester reinforced, self-adhesive, SBS modified bitumen vapour barrier with an aluminium core, with a polyethylene film on the underside and mineral micro granules on the upper surface
- Euroroof Spray SA Primer - a solvent-based primer

The system components have the nominal characteristics given in Table 1.

Table 1 Nominal characteristics

Characteristic (unit)	Euroroof Mono Cap Sheet	Euroroof Self-Adhesive AVCL	Euroroof Torch-On AVCL
Thickness (mm)	4.2	2.0	3.0
Width (m)	1.0	1.0	1.0
Length (m)	8.0	15.0	10.0
Mass per unit area ($kg{\cdot}m^{-2}$)	5.3	2.3	3.7
Weight per roll (kg)	42.4	40	37

Ancillary Items

The Certificate holder recommends the following ancillary items for use with the system, but these materials have not been assessed by the BBA and are outside the scope of this Certificate:

- Monoscreed — a curing screed
- V-Therm VIP — a vacuum insulated panel
- Harmer AV — a range of metal roof outlets
- Alumasc Multi-fix Dual Density Mineral Wool — thermal insulation
- Alumasc Foamglas T4 Ready Board Warm Roof Insulation — foamed glass thermal insulation board
- Skyline — a polyester powder coated aluminium coping, soffit and fascia system
- Modulock — a raised adjustable pedestal system for paving and decking
- Euroroof PU Insulation Adhesive — a single component polyurethane adhesive
- Alumasc GTF PIR thermal insulation — thermal insulation board.

Applications

The system is intended for use in fully or partially bonded flat or pitched roofs with limited access.

The system is intended primarily for use on polyisocyanurate (PIR) insulation and other types of insulation as approved by the Certificate holder.

Definitions for products and applications inspected

The following terms have been defined for the purpose of the Certificate as:

- limited access roof — a roof subjected only to pedestrian traffic for maintenance of the roof covering, cleaning of gutters, etc
- pedestrian access roof – a roof that is not subjected to vehicular traffic
- flat roof — a roof having a minimum finished fall of 1:80
- pitched roof — a roof having a fall in excess of 1:6.

Product assessment – key factors

The system was assessed for the following key factors, and the outcome of the assessments is shown below. Conclusions relating to the Building Regulations apply to the whole of the UK unless otherwise stated.

1 Mechanical resistance and stability

Not applicable.

2 Safety in case of fire

Data were assessed for the following characteristics.

2.1 External fire spread

2.1.1 When tested to CEN/TS 1187 : 2012, Test 4 and classified to EN 13501-5 : 2016, the systems given in Table 2 of this Certificate achieved $B_{ROOF}(t4)$ for slopes below 10°.

Table 2 Tested systems

<table>
<tr><th>Layer</th><th>System 1</th><th>System 2</th><th>System 3</th><th>System 4</th></tr>
<tr><td>Substrate</td><td colspan="4">18 mm OSB (density 600 $kg \cdot m^{-3}$)(1)</td></tr>
<tr><td>Air and vapour control later (AVCL)</td><td colspan="4">2 mm thick, Euroroof Self-Adhesive AVCL</td></tr>
<tr><td>Adhesive</td><td colspan="4">Euroroof PU Insulation Adhesive(1)</td></tr>
<tr><td>Insulation</td><td>30 mm thick Alumasc GTF PIR insulation board(1)</td><td>100 mm thick Alumasc GTF PIR insulation board(1)</td><td>(50 + 100) mm thick Alumasc GTF PIR insulation board(1)</td><td>(100 + 100) mm thick Alumasc GTF PIR insulation board(1)</td></tr>
<tr><td>Adhesive</td><td colspan="4">Euroroof PU Membrane Adhesive</td></tr>
<tr><td>Cap Sheet</td><td colspan="4">4.2 mm Euroroof Mono Cap Sheet</td></tr>
</table>

(1) These components are outside the scope of this Certificate.

2.1.2 On the basis of data assessed, the systems listed in Table 2 will be unrestricted by the documents supporting the national Building Regulations with respect to proximity to a relevant boundary. Restrictions may apply at junctions with compartment walls.

2.1.3 A roof incorporating the system will be similarly unrestricted under the national Building Regulations in protected or inverted roof specifications, including an inorganic covering listed in the Annex of Commission Decision 2000/553/EC.

2.1.4 In Wales and Northern Ireland, when used on flat roofs using a substrate designated in the supporting documents with the surface finishes listed below, a roof will be similarly unrestricted:

- bitumen-bedded stone chippings covering the whole surface to a depth of not less than 12.5 mm
- bitumen-bedded tiles of a non-combustible material
- sand and cement screed
- macadam.

2.1.5 The classification and permissible areas of use of other specifications must be confirmed by reference to the requirements of the documents supporting the national Building Regulations.

2.2 <u>Reaction to fire</u>

2.2.1 The Certificate holder has not declared a reaction to fire classification for the system to BS EN 13501-1 : 2018.

2.2.2 The system will be restricted in use under the documents supporting the national Building Regulations in some cases.

2.2.3 In England, the system, when used in pitches greater than 70°, excluding upstands, must not be used less than 1 m from a relevant boundary, or on residential buildings more than 11 m in height or on other buildings more than 18 m in height. Restrictions apply on assembly and recreation buildings. These constructions must also be included in calculations of unprotected area.

2.2.4 In Wales, the system, when used in pitches greater than 70°, excluding upstands, must not be used less than 1 m from a relevant boundary, or on buildings more than 18 m in height. Restrictions apply on assembly and recreation buildings. These constructions must also be included in calculations of unprotected area.

2.2.5 In Northern Ireland, for systems used on walls or on roofs with pitches greater than 70°, excluding upstands, that do not achieve the minimum Class E reaction to fire classification to BS EN 13501-1 : 2018, designers must seek guidance on the proposed use of the system from the relevant Building Control Body.

2.2.6 In Scotland, the use of the system is unrestricted with respect to building height and proximity to a relevant boundary. However, restrictions on the overall construction may apply, depending on the reaction to fire classification achieved by the build-up, which must be established on a case-by-case basis.

3 Hygiene, health and the environment

Data were assessed for the following characteristics.

3.1 <u>Weathertightness</u>

3.1.1 Results of weathertightness tests are given in Table 3.

Table 3 Weathertightness results

Product assessed	Assessment method	Requirement	Result (Mean)
Euroroof Self-Adhesive AVCL Euroroof Torch-On AVCL	Resistance to Peel from Support (concrete) to MOAT 64	≥25 N·(50 mm)$^{-1}$	Pass Pass
Euroroof Self-Adhesive AVCL Euroroof Torch-On AVCL	Water vapour diffusion - equivalent air layer thickness BS EN 1931 : 2000	Value achieved	21165 m 1789 m
Euroroof Mono Cap Sheet	Shear resistance of joints to NEN EN 12317-1 : 1999	≥ 500 N·(50mm)$^{-1}$	Pass
Built up system: plywood, Euroroof Spray SA Primer, Euroroof Self-Adhesive AVCL Euroroof PU Insulation Adhesive, Alumasc GTF PIR Insulation, Euroroof PU Membrane Adhesive, Euroroof Mono Cap Sheet	Resistance to wind uplift to MOAT 64 : 2001	Value achieved	5.5 kPa
Euroroof Self-Adhesive AVCL Plywood, Euroroof Spray SA Primer Euroroof Self-Adhesive AVCL	Wind uplift to MOAT 64 : 2001	Value achieved	4 kPa

3.1.2 The watertightness of the system was assessed using test data from a representative related product.

3.1.3 On the basis of data assessed, the system, including joints, when completely sealed and consolidated, will adequately resist the passage of moisture to the inside of a building and so satisfy the requirements of the national Building Regulations.

3.1.4 The adhesion of the system is sufficient to resist the effects of wind suction, elevated temperature and thermal shock conditions likely to occur in practice.

3.2 Resistance to mechanical damage

3.2.1 Results of resistance to mechanical damage tests are given in Table 4.

Table 4 Mechanical damage results

Product assessed	Assessment method	Requirement	Result
Euroroof Mono Cap Sheet	Tensile strength to NEN EN 12311-1 : 1999	Value achieved	
	Longitudinal direction		1075 N·(50mm)$^{-1}$
	Transverse direction		950 N·(50mm)$^{-1}$
Euroroof Self-Adhesive AVCL	Longitudinal direction		305 N·(50 mm)$^{-1}$
	Transverse direction		195 N·(50 mm)$^{-1}$
Euroroof Torch-On AVCL	Longitudinal direction		320 N·(50 mm)$^{-1}$
	Transverse direction		195 N·(50 mm)$^{-1}$
Euroroof Mono Cap Sheet	Elongation to NEN EN 12311-1 : 1999	Value achieved	
	Longitudinal direction		43%
	Transverse direction		45%
Euroroof Self-Adhesive AVCL	Longitudinal direction		3%
	Transverse direction		30%
Euroroof Torch-On AVCL	Longitudinal direction		3%
	Transverse direction		30%
Euroroof Mono Cap Sheet	Dynamic indentation to EOTA TR-006 : 2004	Value achieved	
- on aluminium			I_4
- on expanded polystyrene (EPS) insulation			I_3
Euroroof Mono Cap Sheet	Static indentation to EN 12730 : 2015	Value achieved	
- on concrete			20 kg
- on EPS insulation			20 kg
Euroroof Mono Cap Sheet	Resistance to tearing (nail shank) to NEN 12310-1 : 1999	≥ 150 N	
	Longitudinal direction		Pass
	Transverse direction		Pass
Euroroof Self-Adhesive AVCL	Longitudinal direction		Pass
	Transverse direction		Pass
Euroroof Torch-On AVCL	Longitudinal direction		Pass
	Transverse direction		Pass
Euroroof Mono Cap Sheet	Low temperature flexibility to EN 1109 : 2013	≤ -15°C	Pass
Euroroof Self-Adhesive AVCL			Pass
Euroroof Torch-On AVCL			Pass

3.2.2 On the basis of data assessed, the system can accept, without damage, the foot traffic and light concentrated loads associated with installation and maintenance and the effects of minor movement likely to occur in practice, while remaining weathertight.

3.2.3 Where traffic in excess of the example given in section 3.2.2 is envisaged, such as for maintenance of lift equipment, a walkway must be provided. Reasonable care must be taken to avoid puncture by sharp objects or concentrated loads.

4 Safety and accessibility in use

Not applicable.

5 Protection against noise

Not applicable.

6 Energy economy and heat retention

Not applicable.

7 Sustainable use of natural resources

Not applicable.

8 Durability

8.1 The potential mechanisms for degradation and the known performance characteristics of the materials in the system were assessed.

8.2 Specific test data were assessed as given in Table 5.

Table 5 Durability tests

Product assessed	Assessment method	Requirement	Result
Euroroof Mono Cap Sheet	Shear resistance of joints to NEN EN 12317-1 : 1999	≥ 500 N·(50 mm)$^{-1}$	
	Control		Pass
	After 7 days water immersion at 60°C		Pass
Euroroof Mono Cap Sheet	Peel resistance of joints to NEN EN 12316-1 : 1999	≥ 100 N·(50 mm)$^{-1}$	
	Control		Pass
	After 7 days water immersion at 60°C		Pass
Euroroof Mono Cap Sheet Euroroof Self-Adhesive AVCL	Low temperature flexibility to EN 1109 : 2013 Heat aged for 200 days at 70°C	≤ 0°C and a maximum deviation of 15 °C to the initial flexibility at low temperature	Pass Pass
Euroroof Mono Cap Sheet	Heat resistance to EN 1110 : 2010		
	Control	≥ 100°C	Pass
	Heat aged for 200 days at 70°C	≥ 90°C	Pass
Euroroof Mono Cap Sheet Membrane	Resistance to slippage at 45 degrees to MOAT 64 : 2001	< 2mm	Pass
Euroroof Self-Adhesive AVCL	Resistance to Peel from Support (concrete) to MOAT 64 Heat aged 28 days at 80°C	Aged samples should be within 50% of the control and ≥25 N·(50 mm)$^{-1}$	Pass

8.3 <u>Service life</u>

8.3.1 Under normal service conditions, the system will have a life in excess of 25 years, provided it is designed, installed and maintained in accordance with this Certificate and the Certificate holder's instructions.

8.3.2 Localised loss of the mineral surfacing may occur, after some years, in areas where complex detailing of the roof design is incorporated.

PROCESS ASSESSMENT

Information provided by the Certificate holder was assessed for the following factors:

9 Design, installation, workmanship and maintenance

9.1 <u>Design</u>

9.1.1 The design process was assessed by the BBA, and the following requirements apply in order to satisfy the performance assessed in this Certificate.

9.1.2 Decks to which the system is to be applied must comply with the relevant requirements of BS 6229 : 2018, BS 8217 : 2005 and, where appropriate, *NHBC Standards* 2024, Chapter 7.1.

9.1.3 For design purposes of flat roofs, twice the minimum finished fall must be assumed, unless a detailed structural analysis of the roof is available, including overall and local deflection, and direction of falls.

9.1.4 Structural decks to which the system is to be applied must be suitable to transmit the dead and imposed loads experienced in service. Allowance needs to be made for loading deflections to ensure that the free drainage of water is maintained.

9.1.5 Bulk material must not be stored on one area of the roof prior to installation, to ensure that localised overloading does not occur.

9.1.6 Imposed loads, dead loading and wind loads must be calculated by a suitably experienced and competent individual in accordance with BS EN 1991-1-1 : 2002, BS EN 1991-1-3 : 2003 and BS EN 1991-1-4 : 2005, and their UK National Annexes.

9.1.7 The substructure must satisfy the requirements of the relevant clauses of BS 8217 : 2005, and one of the surface finishes described in clause 6.12 of the Code of Practice must be used.

9.1.8 The resistance to wind uplift for warm roofs will be dependent on the cohesive strength of the insulation and the method by which it is secured to the roof deck. This must be taken into account when selecting a suitable insulation material.

9.1.9 The ballast on protected roofs must be of a type that will not be removed or become delocalised owing to wind scour experienced on the roof.

9.1.10 Insulation materials to be used in conjunction with the system must be in accordance with the Certificate holder's instructions and be either:

- as described in the relevant clauses of BS 6229 : 2018, or
- the subject of a current BBA Certificate and be used in accordance with, and within the limitations of, that Certificate.

9.2 Installation

9.2.1 Installation instructions provided by the Certificate holder were assessed and judged to be appropriate and adequate.

9.2.2 Installation must be carried out in accordance with this Certificate and the Certificate holder's instructions, and the relevant clauses of BS 8000-0 : 2014, BS 8000-4 : 1989 and BS 8217 : 2005.

9.2.3 Deck surfaces must be sound, dry and clean, and free from sharp projections such as nail heads and concrete nibs.

9.2.4 The system is laid in conditions normal to roofing work and must not be laid in rain, snow or heavy fog. If the temperature is below 5°C, suitable precautions must be taken against the formation of condensation on the substrate.

9.2.5 At falls in excess of 5° (1:11), precautions against slippage, and requirements for mechanical fixing as required by BS 8217 : 2005, must be observed.

9.2.6 The Euroroof Self-Adhesive AVCL is installed with 75 mm side and 150mm end laps, by removing the release film and firmly pressure rolling the surface to achieve a continuous bond across the full width of the membrane. Ensure that the membrane is accurately aligned, including overlaps, before removing the release film, and that it does not move as the film is removed.

9.2.7 The main field sheet should be pressure rolled with a weighted roller to ensure full contact of the membranes; the addition of hot air may be required to achieve full adhesion in colder conditions.

9.2.8 The side laps (75mm) and head laps (150mm) must be hot air welded and pressure rolled with a long or medium handled 15kg roller. All laps must have a visible bleed of bitumen.

9.2.9 The Euroroof Torch-On AVCL is installed with 100 mm side and 150mm end laps by melting the lower surface by torching and pressing the product down. Care must be taken not to overheat the coating.

9.2.10 The Euroroof Mono Cap Sheet must always be installed with end laps staggered by approximately a quarter of its length from the previous sheet and in such a manner that no counter-seams are made in the direction of outlets.

9.2.11 Corners of membrane which will be laid under the next sheet must be cut at a 45° angle (100 x 100 mm).

9.2.12 The joints, both side and head, must be overlapped by 120 and 150 mm respectively.

9.2.13 For fully bonded applications, bonding is achieved by melting the lower surface by torching and pressing the membrane down. Care must be taken not to overheat the coating.

9.2.14 For adhered applications, Euroroof Mono Cap Sheet must be applied using Euroroof PU Membrane Adhesive at the coverage rate given in Table 6. Laps may be hot air welded or torched and must be pressure rolled with a long handled 15 kg roller from which a bead of compound must flow.

Table 6 Application rates of Euroroof PU Membrane adhesive

Coverage Rate	Coverage per 6 kg tin
Normal	40 – 60 m^2
Roof Perimeters	30 – 40 m^2

9.2.15 Euroroof PU Membrane Adhesive must be applied in 10 mm continuous parallel beads at a rate of 4 beads per metre, at 250 mm centres. The adhesive must not be applied to wet roofs and must be limited to an area that can be covered within 15 minutes of application.

9.2.16 The vertical detail membrane sheet must be applied, ensuring that it is fully bonded to the PIR insulation board and overlapped onto the horizontal area by at least 100 mm.

9.2.17 The height of the vertical details must extend a minimum of 150 mm above the finished surface.

9.2.18 Euroroof PU Membrane Adhesive must not be used at temperatures below 5°C.

9.2.19   The NHBC requires that the system, once installed, is inspected in accordance with *NHBC Standards* 2024, Chapter 7.1, Clause 7.1.11, and undergoes an appropriate integrity test, where required. Any damage to the system assessed in this Certificate must be repaired in accordance with section 9.4 of this Certificate and reinspected, in order to maintain the system's performance.

9.3 Workmanship

Practicability of installation was assessed on the basis of the Certificate holder's information and BS 8217 : 2005. To achieve the performance described in this Certificate, the system must only be installed by contractors who have been trained and approved by the Certificate holder.

9.4 Maintenance and repair

9.4.1 Ongoing satisfactory performance of the system in use requires that it is suitably maintained. The guidance provided by the Certificate holder was assessed by the BBA and found to be appropriate and adequate.

9.4.2 The following requirements apply in order to meet the performance assessed in this Certificate:

9.4.2.1 The system must be the subject of six-monthly inspections and maintenance in accordance with the recommendations in BS 6229 : 2018, Chapter 7, and the manufacturer's own maintenance requirements, where relevant, to ensure continued satisfactory performance.

9.4.2.2 In the event of damage to the waterproof layer, repairs can be carried out by cleaning the area around the damage and applying a patch of the membrane as described in the Certificate holder's instructions.

10 Manufacture

10.1 The production processes for the system have been assessed, and provide assurance that the quality controls are satisfactory according to the following factors:

10.1.1 The manufacturer has provided documented information on the materials, processes, testing and control factors.

10.1.2 The quality control operated over batches of incoming materials has been assessed and deemed appropriate and adequate.

10.1.3 The quality control procedures and testing to be undertaken have been assessed and deemed appropriate and adequate.

10.1.4 The process for management of non-conformities has been assessed and deemed appropriate and adequate.

10.1.5 An audit of each production location was undertaken, and it was confirmed that the production process was in accordance with the documented process, and that equipment has been properly tested and calibrated.

† 10.2 The BBA has undertaken to review the above activities on a regular basis, through a surveillance process, to verify and re-assure that the specifications and quality control operated by the manufacturer are being maintained.

11 Delivery and site handling

11.1 The Certificate holder stated that the system is delivered to site on pallets shrink wrapped in polythene, bearing the product name and bands bearing the Certificate holder's and product names, roll dimensions, production date and batch number.

11.2 Delivery and site handing must be performed in accordance with the Certificate holder's instructions and this Certificate including:

11.2.1 Rolls must be stored upright on a clean, level surface and kept dry, away from excessive heat and under cover.

† ANNEX A – SUPPLEMENTARY INFORMATION

Supporting information in this Annex is relevant to the system but has not formed part of the material assessed for the Certificate.

Construction (Design and Management) Regulations 2015
Construction (Design and Management) Regulations (Northern Ireland) 2016

Information in this Certificate may assist the client, designer (including Principal Designer) and contractor (including Principal Contractor) to address their obligations under these Regulations.

CLP Regulations

The Certificate holder has taken the responsibility of classifying and labelling the system components under the *GB CLP Regulation* and *CLP Regulation (EC) No 1272/2008 - classification, labelling and packaging of substances and mixtures*. Users must refer to the relevant Safety Data Sheet(s).

UKCA marking

The Certificate holder has taken the responsibility of UKCA marking the product in accordance with Designated Standard EN 13707 : 2004, EN 13969 : 2004, EN 14695 : 2010.

CE marking

The Certificate holder has taken the responsibility of CE marking the system components in accordance with harmonised European Standard EN 13707 : 2013.

Bibliography

BS 6229 : 2018 *Flat roofs with continuously supported flexible waterproof coverings — Code of practice*

BS 8217 : 2005 *Reinforced bitumen membranes for roofing — Code of practice*

BS 8000-0 : 2014 *Workmanship on construction sites — Introduction and general principles*
BS 8000-4 : 1989 *Workmanship on building sites — Code of practice for waterproofing*

BS EN 1931 : 2000 *Flexible sheets for waterproofing. Bitumen, plastic and rubber sheets for roof waterproofing. Determination of water vapour transmission properties*

BS EN 1991-1-1 : 2002 *Eurocode 1 : Actions on structures — General actions — Densities, self-weight, imposed loads for buildings*
NA to BS EN 1991-1-1 : 2002 UK National Annex to *Eurocode 1 : Actions on structures — General actions — Densities, self-weight, imposed loads for buildings*
BS EN 1991-1-3 : 2003 + A1 : 2015 Eurocode 1 : Actions on structures — General actions — Snow loads
NA + A2 : 18 to BS EN 1991-1-3 + A1 : 2015 UK National Annex to *Eurocode 1 : Actions on structures — General actions — Snow loads*
BS EN 1991-1-4 : 2005 + A1 : 2010 *Eurocode 1 : Actions on structures — General actions — Wind actions*
NA to BS EN 1991-1-4 : 2005 + A1 : 2010 UK National Annex to *Eurocode 1 : Actions on structures — General actions — Wind actions*

BS EN 13501-1 : 2018 *Fire classification of construction products and building elements — Classification using data from reaction to fire tests*

EN 1109 : 2013 *Flexible sheets for waterproofing — Bitumen sheets for roof waterproofing—Determination of flexibility at low temperature*

EN 1110 : 2010 *Flexible sheets for waterproofing — Bitumen sheets for roof waterproofing. Determination of flow resistance*

EN 12730 : 2015 *Flexible sheets for waterproofing — Bitumen, plastic and rubber sheets for roof waterproofing. Determination of resistance to static loading*

EN 13501-5 : 2016 *Fire classification of construction products and building elements — Classification using data from external fire exposure to roofs tests*

EN 13707:2004 + A2 : 2009 *Flexible sheets for waterproofing — Reinforced bitumen sheets for roof waterproofing — Definitions and characteristics*
EN 13707 : 2013 *Flexible sheets for waterproofing — Reinforced bitumen sheets for roof waterproofing — Definitions and characteristics*

EN 13969 : 2004 + A1 : 2006 *Flexible sheets for waterproofing - Bitumen damp proof sheets including bitumen basement tanking sheets - Definitions and characteristics*

EN 14695 : 2010 *Flexible sheets for waterproofing - Reinforced bitumen sheets for waterproofing of concrete bridge decks and other trafficked areas of concrete - Definitions and characteristics*

EOTA TR-006 : 2004 *Determination of the resistance to dynamic indentation*

CEN/TS 1187 : 2012 *Test methods for external fire exposure to roofs*

MOAT 64 : 2001 *Technical guide for the assessment of roof waterproofing systems made of reinforced APP or SBS polymer modified bitumen sheets*

NEN EN 12310-1 : 1999 *Flexible sheets for waterproofing — Determination of resistance to tearing (nail shank) — Part 1: Bitumen sheets for roof waterproofing*

NEN EN 12311-1 : 1999 - *Flexible sheets for waterproofing — Determination of tensile properties — Part 1 : Bitumen sheets for roof waterproofing*

NEN EN 12316-1 : 1999 *Flexible sheets for waterproofing — Determination of peel resistance of joints — Part 1: Bitumen sheets for roof waterproofing*

NEN EN 12317-1 : 1999 *Flexible sheets for waterproofing bitumen sheets for roof waterproofing — Determination of shear resistance of joints — Part 1: Bitumen sheets for waterproofing*

Conditions of Certificate

Conditions

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